

Let the Sheet Do the Work — Paper D

HARD • NO COMPUTER • NO CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. There is no computer and no calculator: every number is designed to work by hand.
- This paper is about **Google Sheets**. Every formula starts with **=**, arguments are separated by **commas**, and text goes in **quotation marks**.
- Where a question asks for a **value**, write what the cell would show. Where it asks for a **formula**, write it exactly, brackets and commas included.
- The mark for each question is shown in square brackets on the right.
- No list of functions is printed on this paper. You are expected to remember the syntax, including the order of the arguments.
- Several questions describe sheets you may not have seen before. You are not expected to recognise them: apply the ideas you already have.
- A correct value with a wrong formula, or the other way round, earns one mark and not two.

SECTION 1 – READ THE DATA

1. A weather sheet holds July and August rainfall in millimetres for seven cities, with Region in column B.

	A	B	C	D
1	City	Region	Jul	Aug
2	Mumbai	West	840	590
3	Pune	West	180	150
4	Chennai	South	90	115
5	Kochi	South	700	430
6	Kolkata	East	370	340
7	Delhi	North	210	245
8	Jaipur	North	200	230

- (a) A filter is applied: **Jul is greater than or equal to 200** and **Aug is less than 300**. Write the row numbers that remain visible. [2]

- (b) The filter is removed. The **whole table** is sorted by Aug, largest first. Write all seven city names in order, top to bottom. [2]

- (c) Instead, only the Aug cells (D2 to D8) are selected and sorted largest first. [1]
State the Aug value that now sits in the row of Kochi.
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2. A canteen sheet has the Category in column B and the units sold in column C, rows 2 to 6.

	A	B	C
1	Item	Category	Units
2	Chai	Drink	30
3	Samosa	Snack	24
4	Lassi	Drink	18
5	Poha	Meal	12
6	Puff	Snack	16

- (a) A student types =COUNTIF(B2:B6,Drink). State what it shows, and write the corrected formula. [2]
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- (b) Another student types =SUMIF(B2:B6,"Drink"), hoping for the units of drinks sold. State what it shows, and explain why. [2]
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- (c) State the value of =COUNTIF(C2:C6,">=18"). [1]
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SECTION 2 – WRITE AND REPAIR FORMULAS

3. A teacher's grade formula in C2 is meant to show **A** for 80 or more, **B** for 60 to 79, **C** for 40 to 59 and **D** for below 40. The mark is in B2. The formula she typed is:
=IF(B2>=60,"B",IF(B2>=80,"A",IF(B2>=40,"C","D")))

- (a) State what her formula shows for marks of 85, 72, 45 and 30. [2]
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- (b) Write the corrected formula. [3]
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4. A class sheet has the heights of eight pupils in centimetres in B2 to B9. The column was typed by hand.

	A	B
1	Pupil	Height (cm)
2	Ira	152
3	Jai	148
4	Kiran	15
5	Lata	160
6	Mohan	not measured
7	Nisha	155
8	Omkar	
9	Pooja	150

- (a) State the value of =MIN(B2:B9), and say what it tells you about the data. [1]

- (b) State the values of =COUNT(B2:B9) and of =COUNTA(B2:B9). [1]

- (c) State the value of =AVERAGE(B2:B9). [1]

- (d) Write one formula that gives the average of only the valid heights, those of 100 cm or more, and state its value. [2]

SECTION 3 – DESIGN AND DIAGNOSE

5. A sign-up sheet for a school trip has the pupil's House in column B (Red, Blue, Green or Yellow) and Age in column C, for pupils aged 11 to 14.

- (a) Design a data validation rule for column B so that nobody can type a spelling that is not a house. State the criteria and the setting for invalid data. [2]

- (b) Design a rule for column C. State the criteria with the numbers. [1]

- (c) Another student sets the invalid-data option to **Show a warning** instead. State [1]
what happens when someone types **Reds**.

6. Two students wrote the following. Neither statement is correct. The extract shows the sheet they were looking at.

	A	B
1	Subject	Marks
2	Maths	42
3	Science	29
4	Maths	38
5	Science	31
6		
7	Total	=SUM(B2:B5)

- (i) *"I added the rule Number between 0 and 50, Reject the input to a Marks column that already had a 300 typed in it. So the 300 has now been corrected."* [3]
Explain the mistake, then write a correct version.

- (ii) *"I filtered the table to show only Maths. All the Science rows were deleted, so the formula in B7 now shows 80."* [3]
Explain the mistake, then write a correct version.

End of paper. Check that every formula starts with = and that every word inside a formula is in quotation marks.